

OLGA

OLGA Connect

Product, NLP and Technical Blueprint

A vendor-neutral specification for building a professional networking, matchmaking and proximity-aware mobile application.

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Version 1.0 | Concept-to-MVP requirements

Executive summary

OLGA Connect is a consent-first professional network that turns scattered profiles, informal introductions and event proximity into relevant, mutual business conversations. Members describe what they offer and need in everyday language. The product uses semantic NLP to propose useful matches, applies context such as event attendance or opt-in proximity, and opens messaging only after an accepted connection request.

Product principle: Relevance creates the suggestion; member consent creates the connection.

What Release 1 proves

- Members can create credible, searchable professional profiles without exposing contact details.
- Free-text needs and offers can be converted into meaningful match candidates, including paraphrases and related terminology.
- Nearby and event-based context can improve timing without continuously exposing a member location.
- A request/accept workflow prevents unsolicited messaging and creates a clean privacy boundary.
- The system produces explainable match reasons so the NLP is useful, inspectable and safe to improve.

Success measures

Measure	Release 1 target	Why it matters
Profile completion	70% of activated members	Matching quality depends on usable intent and profile data.
Match-to-request rate	>= 10% of viewed matches	Tests whether suggestions feel relevant.
Request acceptance	>= 35% of requests	Signals mutual usefulness and trust.
Time to first relevant match	< 10 minutes after profile completion	Makes the product immediately valuable.
NLP precision review	>= 80% positive in labelled test set	Establishes a measurable learning loop.

1. Product vision and boundaries

Vision

Make it easy for professionals in the same ecosystem to discover potential collaborators, suppliers, customers and partners at the moment a conversation is most useful.

In scope for Release 1

- Mobile member app for iOS and Android from one shared codebase.
- Email or mobile-based sign-up, profile verification workflow and role-based administration.
- Profiles, free-text needs/offers, semantic match recommendations and ranked match feed.
- Opt-in event check-in and coarse proximity awareness.
- Connection requests, mutual acceptance, one-to-one chat and controlled file sharing.
- Admin console for member moderation, event configuration, content review and match-quality evaluation.

Explicitly deferred

- Public web directory, payments, marketplace transactions and enterprise CRM/ERP integrations.
- Automatic scraping or two-way synchronization from third-party social networks.
- Background precise-location tracking, contact export and direct messaging before consent.

- Automated decision-making that blocks participation, prices services or makes high-impact decisions.

2. Users, jobs and end-to-end journey

User	Primary job	Information needed
Member	Find a relevant professional connection	Role, organization, offer, need, market, event context.
Event participant	Prioritize valuable conversations onsite	Event presence, coarse proximity, shared context, match reason.
Community administrator	Keep the network trusted and healthy	Verification status, reports, moderation controls, quality metrics.
NLP evaluator	Improve recommendation quality	Labelled match pairs, feedback reasons, model/version telemetry.

Core member journey

1. Register using a verified email address or mobile number; agree to community and location consent terms.
2. Complete profile: role, organization, sector, geography, professional summary, offers and needs.
3. Review suggested matches ranked by semantic relevance and context. Each card explains the top matching reasons.
4. Optionally activate Live Mode at a selected event or venue. The app checks coarse proximity only while Live Mode is active.
5. Send a connection request with an optional short note. The recipient can accept, decline, block or report.
6. After acceptance, unlock chat and controlled file sharing. Both parties can disconnect or revoke sharing later.

Key screen states

State	Required response
First use	Guide profile completion and collect consent progressively.
No matches	Explain which profile fields or intent details would improve suggestions.
Location disabled	Continue semantic matching; explain how to enable Live Mode later.
Network unavailable	Queue safe actions, show sync status and avoid duplicate requests.
Permission denied	Respect denial; never loop prompts; provide a settings path.
Match unavailable	Hide sensitive rationale and offer a non-technical fallback message.

3. Functional requirements

Identity and profiles

ID	Requirement	Priority
FR-01	The system shall support email or mobile registration, account verification, sign-in and passwordless recovery.	Must
FR-02	A member shall manage profile, role, organization, sectors, operating geography, offers, needs and visibility preferences.	Must
FR-03	An administrator shall review, approve, suspend or anonymize a member account with an audit trail.	Must
FR-04	The member shall choose which profile fields are visible before a connection is accepted.	Must

Matching and recommendations

ID	Requirement	Priority
FR-05	The system shall accept natural-language offer and need statements and store both original text and normalized NLP representation.	Must
FR-06	The system shall generate candidate pairs using vector similarity and rank them with configurable business rules.	Must
FR-07	Every displayed recommendation shall include a human-readable explanation and an option to mark it useful, not useful or inappropriate.	Must
FR-08	The system shall prevent self-matches, blocked pairs, already-connected pairs and policy-ineligible profiles.	Must
FR-09	Administrators shall configure ranking thresholds, event scope and recommendation suppression rules.	Should

Consent, proximity and events

ID	Requirement	Priority
FR-10	A member shall explicitly opt in to Live Mode for a defined event or venue and may turn it off at any time.	Must
FR-11	The system shall use coarse location or event check-in to determine presence; precise coordinates shall not be displayed to other members.	Must
FR-12	A nearby alert shall only be generated for a high-confidence semantic match that satisfies consent and connection rules.	Must
FR-13	The system shall rate-limit alerts and provide quiet hours to prevent notification fatigue.	Must

Connections, chat and sharing

ID	Requirement	Priority
FR-14	A member shall send, accept, decline, withdraw, block or report a connection request.	Must
FR-15	One-to-one chat shall become available only after mutual connection acceptance.	Must
FR-16	Connected members shall exchange messages and files subject to file type, size and malware-scanning policies.	Must
FR-17	The system shall retain moderation metadata and permit message/report review only for authorized roles under a documented policy.	Should

Administration and analytics

ID	Requirement	Priority
FR-18	Administrators shall manage events, member verification, reports, blocked-content rules and consent-support requests.	Must
FR-19	The system shall capture privacy-safe product analytics for activation, matching, requests, acceptance, chat and feedback.	Must
FR-20	The NLP evaluator shall export de-identified labelled match pairs and view model, threshold and feedback metrics.	Should

4. NLP and semantic matching design

Learning objective

Build the matching system as an observable pipeline, not a black box. Begin with embeddings plus rules, collect feedback, then iterate through evaluation before attempting fine-tuning or complex autonomous agents.

Layer	Design	Output
Intent capture	Separate free-text fields for “I offer” and “I need”; structured fields for sector, geography and role.	Source intent record.
Normalization	Language detection, safe text cleanup, optional term extraction and redaction of obvious personal data.	Normalized text plus metadata.
Embedding	Create vector representations for offer and need statements using an approved embedding model.	Offer and need vectors.
Candidate retrieval	Search for reciprocal candidates: A need against B offer and B need against A offer.	Top-K candidate pairs.
Ranking	Blend semantic score, reciprocity, sector/geography fit, freshness, event context and safety filters.	Ranked match score.
Explanation	Generate templated reasons from the contributing signals; do not expose raw vectors or sensitive data.	Match reasons.
Feedback loop	Collect useful/not-useful labels, monitor drift and maintain a gold evaluation set.	Metrics and training evidence.

NLP scoring and evaluation

Illustrative ranking formula

Example only: $\text{final_score} = 0.50 \times \text{reciprocal semantic relevance} + 0.15 \times \text{sector fit} + 0.10 \times \text{geography fit} + 0.10 \times \text{intent freshness} + 0.10 \times \text{event context} + 0.05 \times \text{profile completeness}$. Policy filters run before scoring and can suppress any candidate.

The semantic term contains two directional scores: how well Member B offers what Member A needs, and how well Member A offers what Member B needs. Use the lower or harmonic mean when mutual benefit is essential; use a weighted directional score when a buyer-supplier flow is valid.

Evaluation requirements

- Create a labelled set of at least 200 candidate pairs with categories: strong match, plausible match, weak match, no match and unsafe/ineligible.
- Split labels by member or organization so near-duplicate intent does not leak between validation and test sets.
- Track Precision@5, Recall@20, reciprocal-match precision, acceptance rate and the rate of “not useful” feedback.
- Review false positives by category: vocabulary ambiguity, missing profile context, geography mismatch, stale intent, policy error or model limitation.
- Version every embedding model, ranking configuration and evaluation dataset; enable rollback when quality degrades.

Prompting and generative AI guardrails

- Use deterministic templates for match explanations in the MVP. If a language model is added, restrict it to summarization or draft assistance and validate output before display.
- Do not send raw private messages, exact locations or unneeded personal data to an external model service.
- Treat generated text as untrusted: filter unsafe content, log only approved telemetry and never let generated text bypass consent or moderation rules.

5. Reference architecture

The implementation should use interchangeable components so the learning project can begin locally and graduate to managed cloud services without rewriting product logic.

Component	Responsibilities	Implementation guidance
Mobile app	Onboarding, profile, match feed, requests, chat, consent and notifications.	Cross-platform TypeScript application; local encrypted storage for tokens only.
API and identity	Authentication, authorization, profile APIs, consent records and rate limits.	REST or GraphQL edge with role-based access control and short-lived tokens.
Core database	Transactional member, profile, request, event, consent and audit data.	Relational database; location extension for coarse geo queries.
Vector store	Embedding persistence and nearest-neighbour retrieval.	Vector extension in the relational database for MVP; separate service only when scale requires it.
Matching worker	Normalization, embedding, candidate retrieval, ranking and explanation generation.	Asynchronous job worker with idempotent runs and versioned outputs.
Realtime service	Chat delivery, presence and request updates.	Authenticated WebSocket gateway with durable message persistence.
Object storage	Private file uploads, malware scan handoff and expiring downloads.	Encrypted object storage with signed URLs and retention controls.
Admin console	Moderation, event configuration, analytics and NLP evaluation review.	Web app protected by strong administrator authentication.

Architecture flow and data model

Flow	Control point	Result
1. Member enters offer/need	Validation and consent check	Intent record saved.
2. Worker normalizes and embeds	PII minimization, version tag	Vectors stored and indexed.
3. Candidate retrieval and ranking	Eligibility, blocks, threshold	Safe ranked candidates.
4. Match card delivery	Explanation policy, exposure settings	Member sees a controlled recommendation.
5. Request and acceptance	Mutual-consent gate	Chat capability provisioned.
6. Feedback and analytics	De-identification and retention policy	NLP evaluation signals retained.

Core entities

Entity	Important fields	Relationships
Member	id, status, verified_at, visibility, locale	One profile; many intents, consents, requests and messages.

Entity	Important fields	Relationships
Profile	member_id, role, organization, sectors, geography, summary	Used for eligibility and explanation.
Intent	id, member_id, type, original_text, normalized_text, freshness	One or more embeddings; type = offer or need.
Embedding	intent_id, model_version, vector, created_at	Retrieved for semantic candidate search.
MatchCandidate	pair_id, scores, reasons, policy_status, rank_version	Ephemeral or time-bounded recommendation record.
Consent	member_id, purpose, status, granted_at, revoked_at	Authoritative evidence for Live Mode and analytics uses.
EventPresence	member_id, event_id, coarse_cell, active_until	Short-retention context for nearby matching.
ConnectionRequest	sender_id, recipient_id, status, note, timestamps	Creates conversation only when accepted.
Conversation / Message	participant ids, body, attachment metadata, moderation state	Visible only to accepted connection pair.

API surface (illustrative)

Method	Endpoint	Purpose
POST	/auth/register, /auth/verify, /auth/session	Account creation and secure session lifecycle.
GET/PATCH	/me/profile	Read and update member profile and visibility.
POST/GET	/intents, /matches	Submit intent and retrieve paginated recommendations.
POST	/matches/{id}/feedback	Capture useful, not useful or report feedback.
POST/GET	/events/{id}/live-mode	Activate/deactivate bounded event presence.
POST/PATCH	/connection-requests, /connection-requests/{id}	Send, accept, decline or withdraw requests.
GET/POST	/conversations, /conversations/{id}/messages	Retrieve and send messages after authorization.
GET/POST	/admin/*	Moderation, events, evaluation and audit functions.

6. Security, privacy and compliance requirements

Area	Requirement
Consent	Record purpose-specific, revocable consent for location, event presence, notifications and analytics. Default all optional collection to off.
Location	Store coarse cells or event-presence signals wherever feasible. Do not expose another member's exact location, history or arrival time.
Access control	Use least privilege, role-based authorization, server-side checks on every object and administrator MFA.
Data protection	Encrypt data in transit and at rest; manage secrets outside source code; rotate credentials and short-lived access tokens.
Files and chat	Scan uploads, restrict MIME types and size, issue expiring download URLs, and apply retention/deletion rules.
Abuse protection	Rate-limit authentication, matching, requests and messages; support blocking, reporting and abuse-review workflows.
Auditability	Log sensitive admin actions, consent changes, export attempts, policy suppressions and model/ranking versions.
Data rights	Provide account deletion, data export where lawful, correction and consent withdrawal flows with verifiable completion.

Threats to test

- Enumeration of member identities or profiles through search, nearby alerts or predictable IDs.
- Unauthorized access to chat, attachments or administrative exports.
- Location inference from repeated alerts or event-presence timestamps.
- Prompt injection or data leakage if generative AI services are introduced.
- Spam, harassment, malicious attachments and automated request flooding.

7. Non-functional requirements

Category	Requirement
Performance	Match feed p95 response under 1.5 seconds for cached candidates; request and message acknowledgement p95 under 500 ms under target load.
Availability	Design for graceful degradation: semantic recommendations and chat may degrade independently; consent and access controls must never fail open.
Scalability	Queue embedding and ranking work; partition analytics from transactional paths; load-test separately for chat and candidate generation.
Reliability	Idempotency keys for writes, retry-safe jobs, backups, restore drills and dead-letter queues for failed background work.
Accessibility	Mobile interfaces meet WCAG 2.2 AA intent: contrast, screen-reader labels, focus order, dynamic text and non-color status cues.
Observability	Structured logs, metrics, traces, security alerts and dashboards for match quality, worker latency, delivery failures and consent errors.
Localization	Keep user-visible strings externalized; support locale-aware dates, geography and multilingual intent processing as a phased capability.

8. Delivery plan and acceptance criteria

Stage	Outcome	Acceptance evidence
1. Discovery and UX	Validated personas, consent language, user flows and clickable prototype.	Stakeholder walkthrough includes empty, loading, error and permission-denied states.
2. Data and NLP foundation	Schema, intent pipeline, embeddings, labelled evaluation set and baseline ranking.	Offline evaluation report with model/version, metrics and error analysis.
3. Core mobile experience	Registration, profile, intent, match feed, request flow and chat.	Automated tests plus scripted demo across iOS and Android target devices.
4. Live Mode and administration	Event check-in, bounded proximity alerts, moderation and admin console.	Consent revocation, location-denial and alert-rate tests pass.
5. Hardening and pilot	Security review, performance tests, observability and small cohort pilot.	Pilot feedback reviewed; severity-1 defects closed; rollback plan rehearsed.

Definition of done for Release 1

- All Must functional requirements are demonstrated in a test environment and documented with test cases.
- A labelled NLP evaluation set and repeatable evaluation script exist; baseline quality and known failure modes are recorded.
- No direct message can be initiated before accepted connection status; blocked pairs never appear as matches.
- Live Mode can be enabled, disabled and revoked immediately; coarse location data adheres to configured retention rules.

- Critical security tests pass; secrets are not in source control; backup and restore procedure is documented.
- A pilot administrator can moderate members, reports and events without developer intervention.

9. Backlog: post-MVP opportunities

Theme	Potential capability	Why later
Trust	Recommendations, endorsements, verified credentials and reputation signals.	Needs mature moderation policy and careful incentive design.
Events	Event discovery, agenda context and organizer tools.	Requires partnerships, event data quality and dedicated operations.
Commercial	Structured quote requests, marketplace workflows and payments.	Adds legal, financial and dispute-resolution obligations.
Enterprise	CRM/ERP integrations, team accounts and procurement workflows.	Needs stable APIs, tenant isolation and enterprise security controls.
NLP maturity	Domain adaptation, multilingual matching, re-ranking and active learning.	Should follow evidence from labelled feedback rather than precede it.

10. Assumptions, risks and decisions to make

Area	Assumption / decision	Risk if unresolved
Community	OLGA defines the initial member segment and verification standard.	Low-quality or untrusted profiles weaken every match.
NLP domain	The first pilot has enough representative offer/need examples to label.	Generic semantic results may feel broad or inconsistent.
Privacy	OLGA approves consent copy, retention windows and moderation access policy.	Location and messaging features can create avoidable trust and compliance issues.
Event operations	Events have a reliable check-in or bounded venue configuration.	Nearby signals become noisy, intrusive or technically fragile.
Pilot	A small group of real members agrees to test and provide feedback.	NLP quality cannot improve from synthetic examples alone.
Ownership	OLGA maintains source code, data ownership, service accounts and deployment access from the beginning.	Handover and long-term maintainability become difficult.

Recommended first build sequence

7. Create a local dataset of anonymized professional offers and needs, then label representative reciprocal matches.
8. Implement profile and intent entry before location or chat; prove match relevance offline first.
9. Add vector retrieval, explainable scoring and feedback controls; instrument every recommendation.
10. Introduce requests and accepted-only chat. Test access-control boundaries thoroughly.
11. Pilot event check-in with coarse presence, rate-limited alerts and simple opt-in controls.
12. Use pilot feedback to revise taxonomy, threshold and ranking weights before adding advanced AI or commercial features.

Build note: This is a technical product specification, not a vendor proposal. Specific cloud providers, identity platforms, model providers and commercial estimates should be selected separately after architecture, privacy and pilot decisions are made.